

Pharmacy Management System

LLD

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1. Document Purpose

The purpose of this document is to describe the Low-Level Design (LLD) and key system components for the Pharmacy Management System developed using ASP.NET Core and Entity Framework Core with SQL Server. It defines the data flow, entities, and interactions between users (Admin and Doctor) and the system. The document serves as a blueprint for developers to build, integrate, and test the system efficiently.

2. Project Background and Objectives

2.1 Background

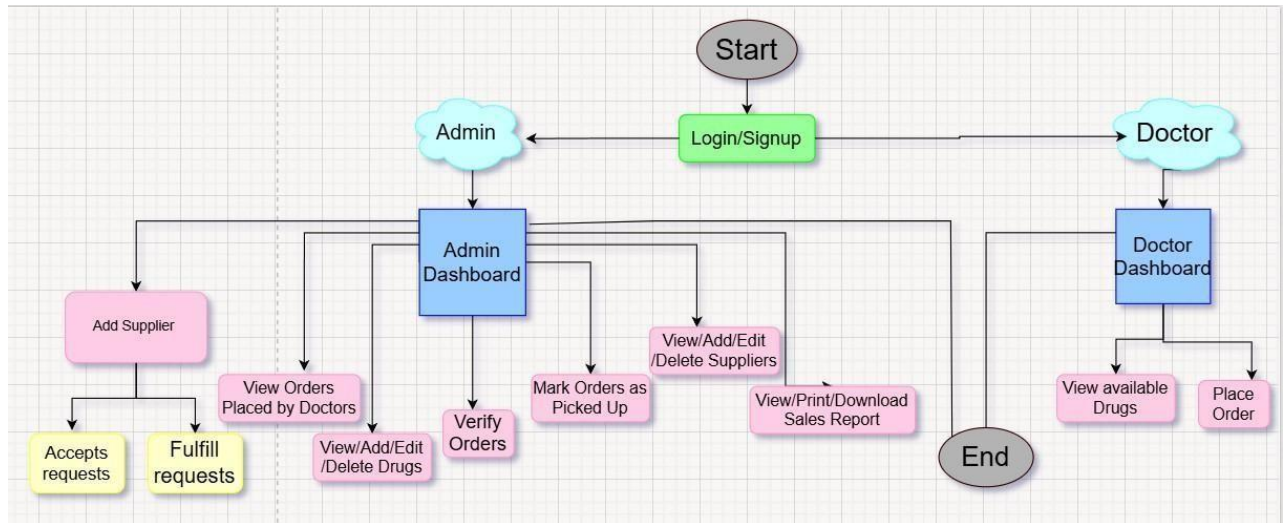
Pharmacies play a vital role in healthcare management by ensuring that patients and doctors have access to the necessary drugs. Managing inventory, verifying doctor orders, and maintaining supplier information manually is time-consuming and prone to human errors.

The Pharmacy Management System automates these operations, ensuring accurate data handling, efficient stock management, and seamless order tracking.

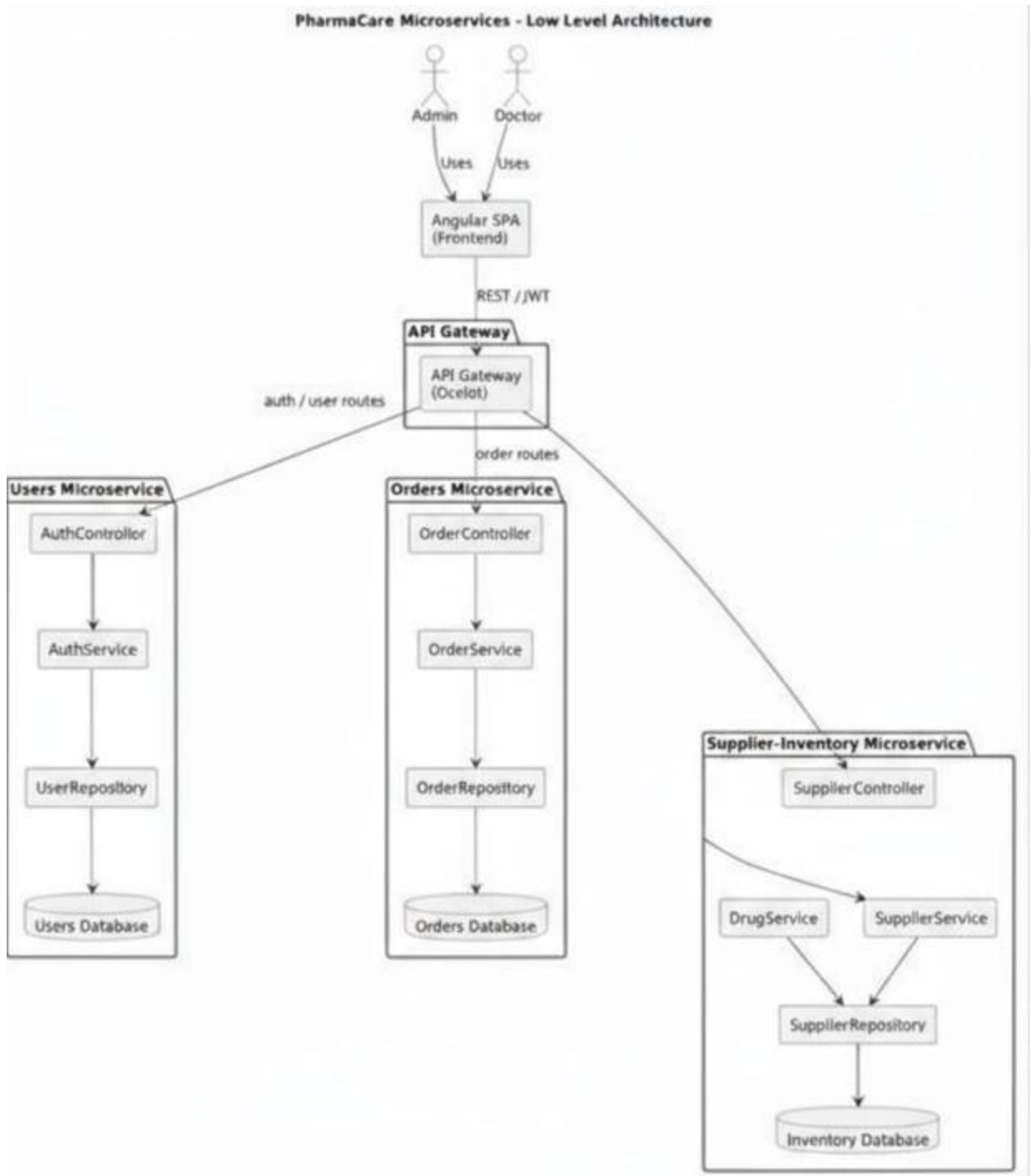
2.2 Objectives

- To create a secure and efficient backend system for managing pharmacy operations.
- To allow Admin users to manage drugs, suppliers, and doctor orders digitally.
- To allow Doctor users to view available drugs and purchase (order) them online.
- To generate sales reports automatically for admins to track revenue and performance.
- To ensure data consistency, security, and easy scalability through .NET and SQL Server.

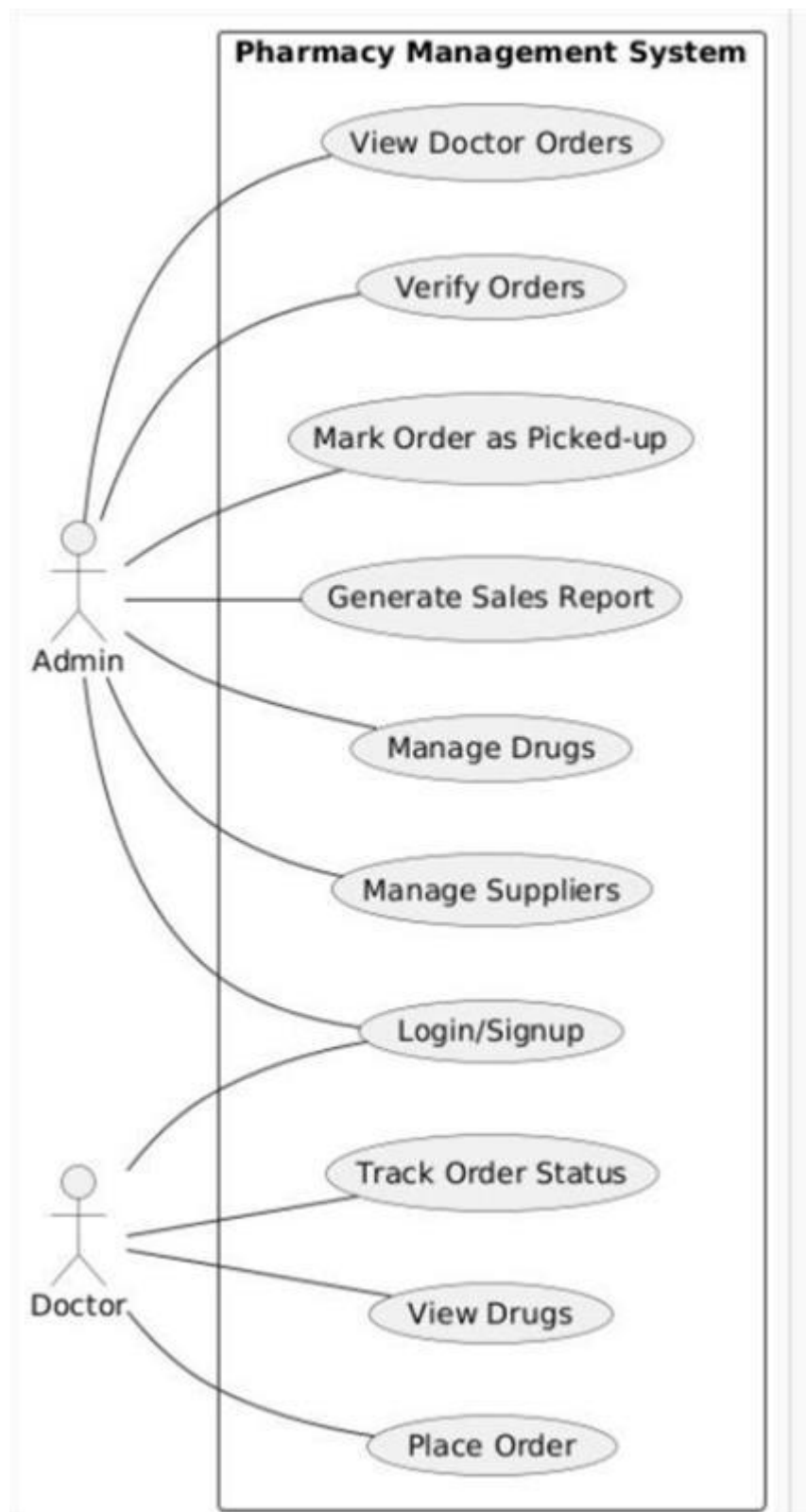
3. Flow Diagram



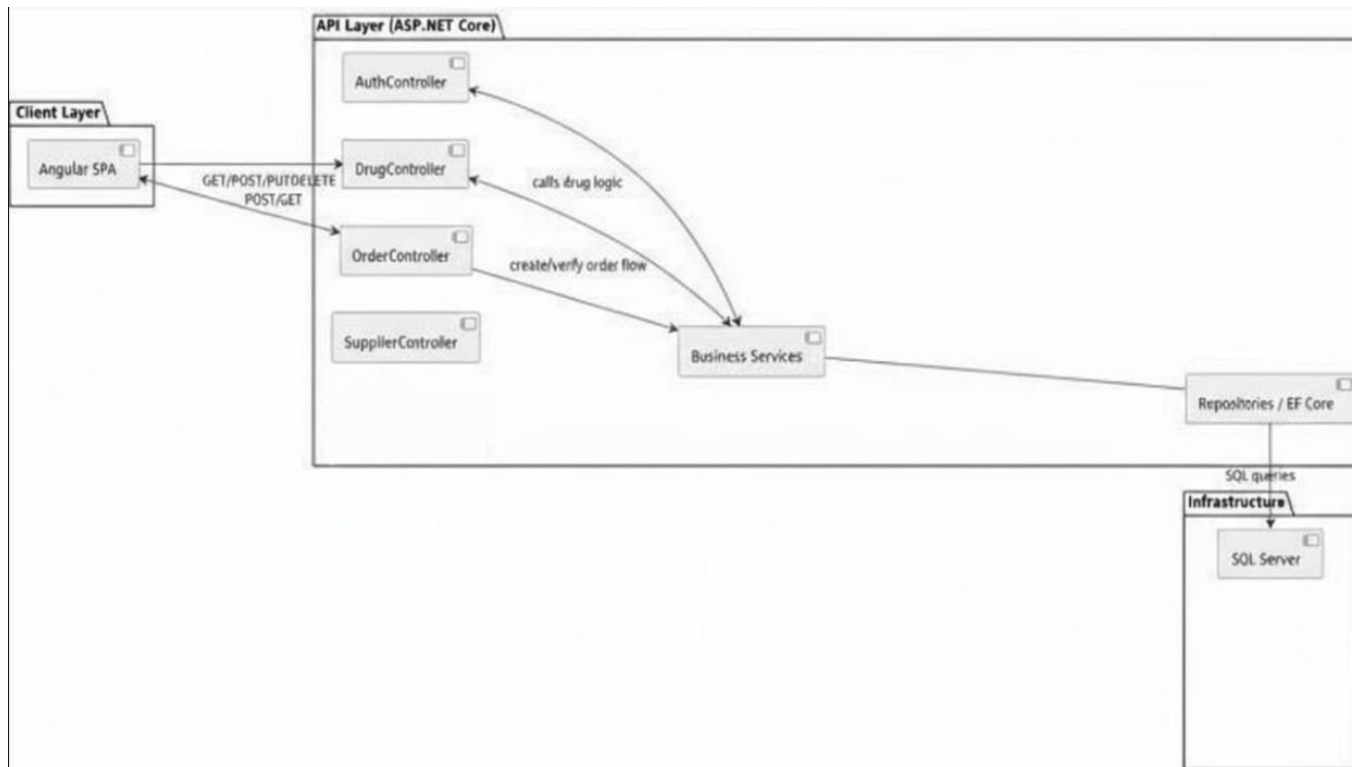
4. Architecture Diagram



5. Use Case Diagram



6. Solution Diagram



7. Database Diagram

Database Schema

Roles		
PK	Id	INTEGER(12)
	Name	VARCHAR(10)

User		
PK	Id	INTEGER(10)
	Role	VARCHAR(20)
	LastLoginDate	DATETIME(20)
	IsActive	BOOLEAN

Supplier		
PK	SupplierId	INTEGER(10)
	Name	VARCHAR(50)
	EmailAddress	VARCHAR(15)
	PhoneNumber	VARCHAR(13)

Sales		
PK	SalesId	INTEGER(10)
	Date	DATETIME(20)
	TotalSales	DECIMAL(10)
	PaymentMethod	VARCHAR(25)
FK	DrugId	INTEGER(10)

Drugs		
PK	DrugId	INTEGER(10)
	Name	VARCHAR(20)
	Manufacturer	VARCHAR(20)
	Price	DECIMAL(12)
	Stock	INTEGER(10)
	StorageInstructions	VARCHAR(50)
	IsPrescriptionRequired	BOOLEAN

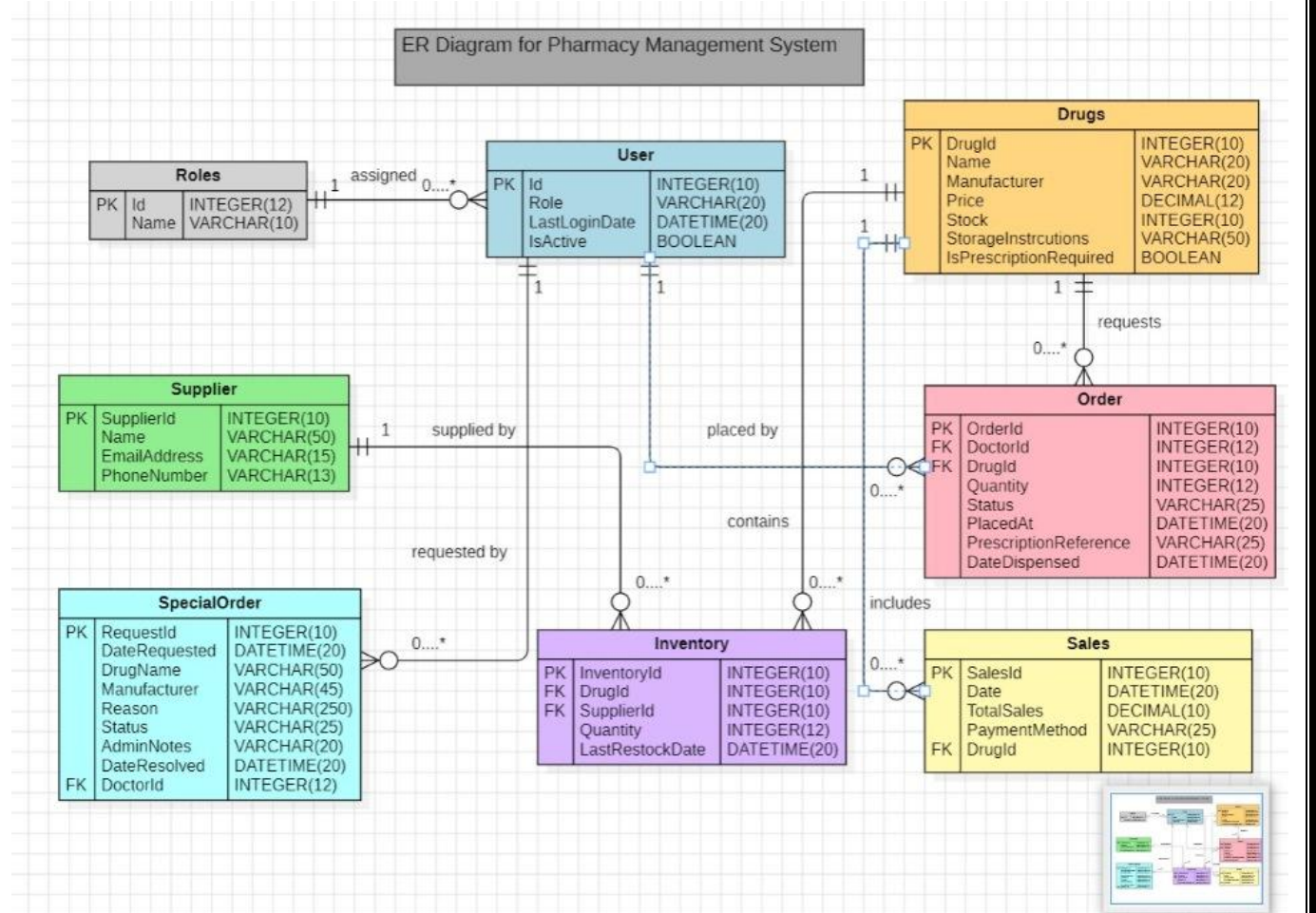
SpecialOrder		
PK	RequestId	INTEGER(10)
	DateRequested	DATETIME(20)
	DrugName	VARCHAR(50)
	Manufacturer	VARCHAR(45)
	Reason	VARCHAR(250)
	Status	VARCHAR(25)
	AdminNotes	VARCHAR(20)
	DateResolved	DATETIME(20)
FK	DoctorId	INTEGER(12)

Inventory		
PK	InventoryId	INTEGER(10)
FK	DrugId	INTEGER(10)
FK	SupplierId	INTEGER(10)
	Quantity	INTEGER(12)
	LastRestockDate	DATETIME(20)

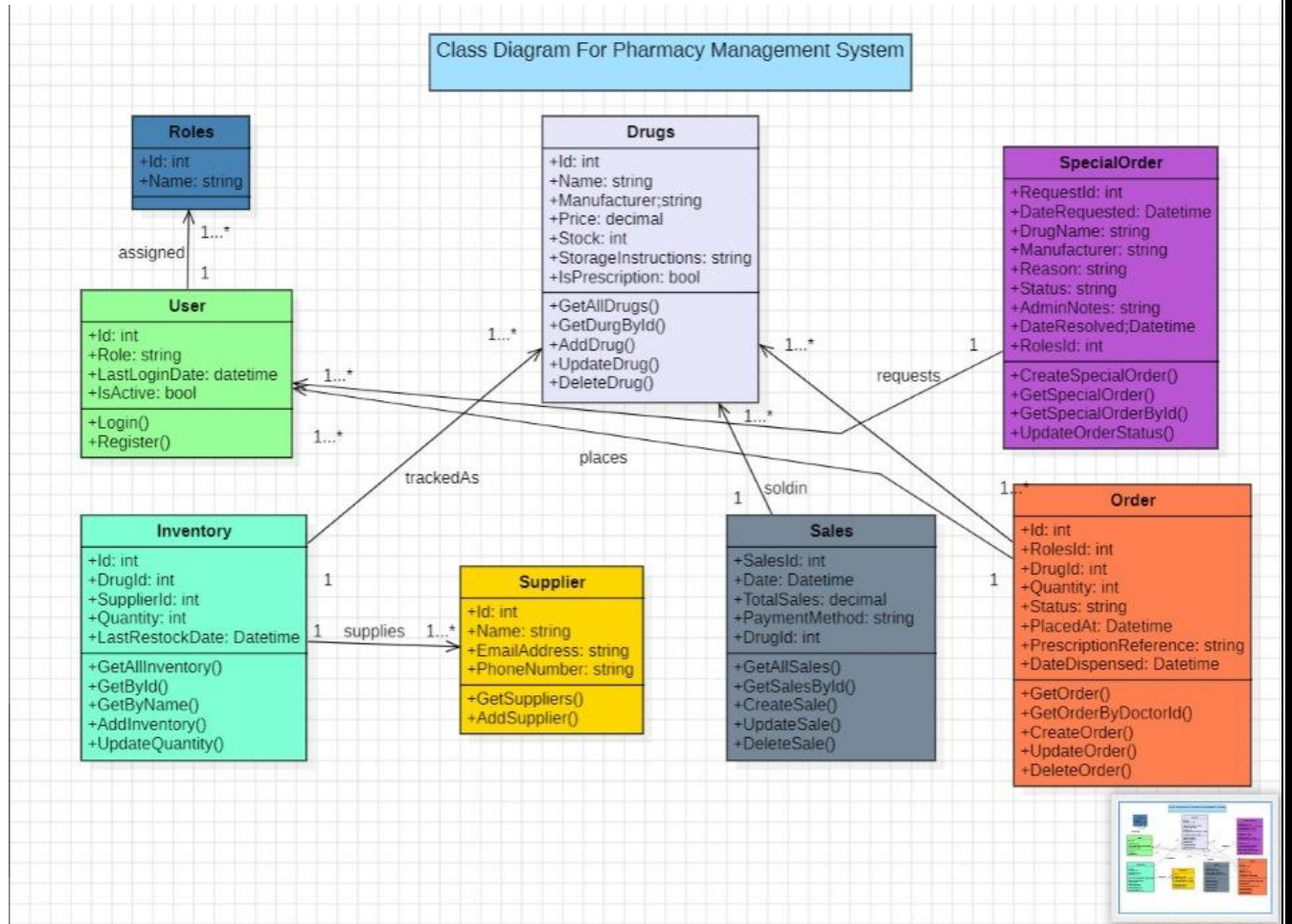
Order		
PK	OrderId	INTEGER(10)
FK	DoctorId	INTEGER(12)
FK	DrugId	INTEGER(10)
	Quantity	INTEGER(12)
	Status	VARCHAR(25)
	PlacedAt	DATETIME(20)
	PrescriptionReference	VARCHAR(25)
	DateDispensed	DATETIME(20)



8. ER Diagram



9. Class Diagram



10. User Requirements

10.1 Hardware Requirements

Minimum:

- i5 Processor: A mid-range Intel CPU suitable for basic operations.
- 8 GB RAM: Enough for running the application and browser smoothly.
- 256 GB SSD: Fast storage for quick boot and load times.

Recommended:

- i7 Processor: Faster performance, especially for multitasking.
- 16 GB RAM: Ideal for smoother performance with multiple apps running.

10.2 Software Requirements

Operating System:

- Windows 10 / 11: Compatible with .NET and development tools.

Browser:

- Chrome / Edge: Modern browsers that support Angular-based frontends.

SDK:

- .NET 8 SDK: Software Development Kit for building and running .NET Core applications.

Database:

- Microsoft SQL Server: A relational database system to store pharmacy data like inventory, prescriptions, and customer records.

Tools:

- VS Code (Visual Studio Code): A lightweight, open-source code editor with support for .NET, Angular, and many extensions.
- SQL Server Management Studio (SSMS): A GUI tool to manage SQL Server databases.

- Angular CLI: Command Line Interface for creating and managing Angular applications.

11 Developer Requirements

11.1 Hardware

- Developer Machines: Should have a powerful CPU and sufficient RAM (ideally i7, 16 GB RAM or more) to handle builds, debugging, and running both backend and frontend services simultaneously.

11.2 Software

- .NET 8 SDK: Required to build and run the backend Web API.
- Visual Studio Code: Used for writing and debugging code.
- SQL Server Developer Edition: Full-featured free edition of SQL Server for development and testing.
- Swagger: Tool for API documentation and testing. It provides a UI to test API endpoints.
- Entity Framework Core: An Object-Relational Mapper (ORM) for .NET to interact with the database using C# objects.
- Angular 15+: Framework for building the frontend (user interface) of the application.

11.3 Technology

Backend: ASP.NET Core Web API

- A framework for building RESTful APIs using C#.
- Handles business logic, data processing, and communication with the database.

Frontend: Angular

- A TypeScript-based framework for building dynamic web applications.
- Provides a responsive and interactive user interface for pharmacists and staff.

Database: SQL Server

- Stores structured data like medicine inventory, sales records, prescriptions, and customer details.

Documentation: Swagger UI

- Automatically generates interactive API documentation.
- Allows developers and testers to explore and test endpoints directly from the browser.

Authentication: JWT Tokens (JSON Web Tokens)

- A secure way to handle user authentication.
- Tokens are issued upon login and used to authorize API requests.